MD. ARIFUR RAHMAN MULLA

Rajshahi, Bangladesh

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RESEARCH INTERESTS

GIS & Remote Sensing | Transportation Planning & Engineering | Land Use Planning | Land Readjustment | Climate-Resilient Urban Development | Disaster Risk & Drought Assessment | Urban Analytics & Planning Policy

EDUCATION

Rajshahi University of Engineering & Technology (RUET) *Dec 2023 - Present*

Bachelor of Urban & Regional Planning | Department of Urban & Regional Planning *Rajshahi, Bangladesh*

- **Student ID:** 2207014 |
- **Academic focus:** GIS, remote sensing, transportation planning, land use planning, land economics, climate resilience, environmental planning, and research methodology.
- **Relevant coursework:** GIS & Remote Sensing, Transportation Planning, Land Use Planning, Land Economics, Environmental Planning, Natural Hazards & Disaster Management, Surveying, Statistics, Planning Studio, and Research Methods.

TECHNICAL SKILLS

Geospatial & Remote Sensing: ArcGIS Pro, ArcMap 10.8, QGIS, Google Earth Engine, Google Earth Pro
Planning, Design & Mapping: AutoCAD, SketchUp, land-use mapping, road/network analysis, urban layout preparation, map visualization
Other Applications: Microsoft Word, Excel, PowerPoint; SPSS and Adobe tools (basic/familiar)
Languages: Bangla (Native), English (Academic/working proficiency)

ACADEMIC & RESEARCH EXPERIENCE

Natural Capital and Drought Impact Assessment in Tanore Upazila *2026*

Rajshahi, Bangladesh

Undergraduate Research Project | RUET

- Assessed drought impacts on natural capital using GIS and remote sensing indicators such as NDVI, LST, NDWI, rainfall/SPI, soil moisture, groundwater depth, crop productivity, land degradation, and vegetation buffering.
- Prepared a composite vulnerability workflow using normalization, PCA-based weighting, index mapping, drought severity-duration interaction, and vulnerability class interpretation.
- Designed survey and KII tools and prepared maps, methodology diagrams, report sections, results slides, and adaptation recommendations for climate-resilient planning.

Integrated Land Readjustment Barrier-Solution Framework and Blue-Green Prototype Plan *2026*

Rajshahi, Bangladesh

Undergraduate Research Project | Katakhali Paurashava Case Study

- Developed a policy-oriented land readjustment framework by analyzing stakeholder barriers, landowner perceptions, occupation-willingness relationships, and institutional implementation challenges.
- Applied Relative Importance Index (RII) analysis for barrier ranking and prepared survey/KII instruments for landowner and professional stakeholder responses.
- Prepared a blue-green prototype plan for a 160-acre net developable area, including road hierarchy, residential and commercial plots, community facilities, reserved plots, open spaces, and water bodies.
- Developed cost estimation and cost-recovery logic with contribution-ratio justification, reserve plot sales assumptions, and implementation recommendations.

Urban Growth, LULC and Thermal Anomaly Modelling Research *2026*

RUET

Academic Paper Preparation | GIS, ABM and Random Forest

- Prepared an academic paper using GIS/remote sensing, agent-based modelling (ABM), Random Forest, land-use/land-cover change, and land surface temperature analysis for urban planning decision support.
- Developed abstract, introduction, methodology flow diagram, results discussion, limitations, conclusion, references, and editable Word-based figures in academic format.

LEADERSHIP & VOLUNTEERING EXPERIENCE

RUET GIS Club *2025 - Present*

Executive Member *Rajshahi, Bangladesh*

- Supported the organization of GEOPLAN 1.0, a GIS-based event with academic competitions, participant coordination, certificate preparation, and formal communication.
- Contributed to GIS learning activities, workshop planning, map-based project discussion, and promotion of geospatial tools among Urban & Regional Planning students.
- Prepared event documents, apology/certificate communication drafts, reports, and visual materials for club activities.